Library Database Management System

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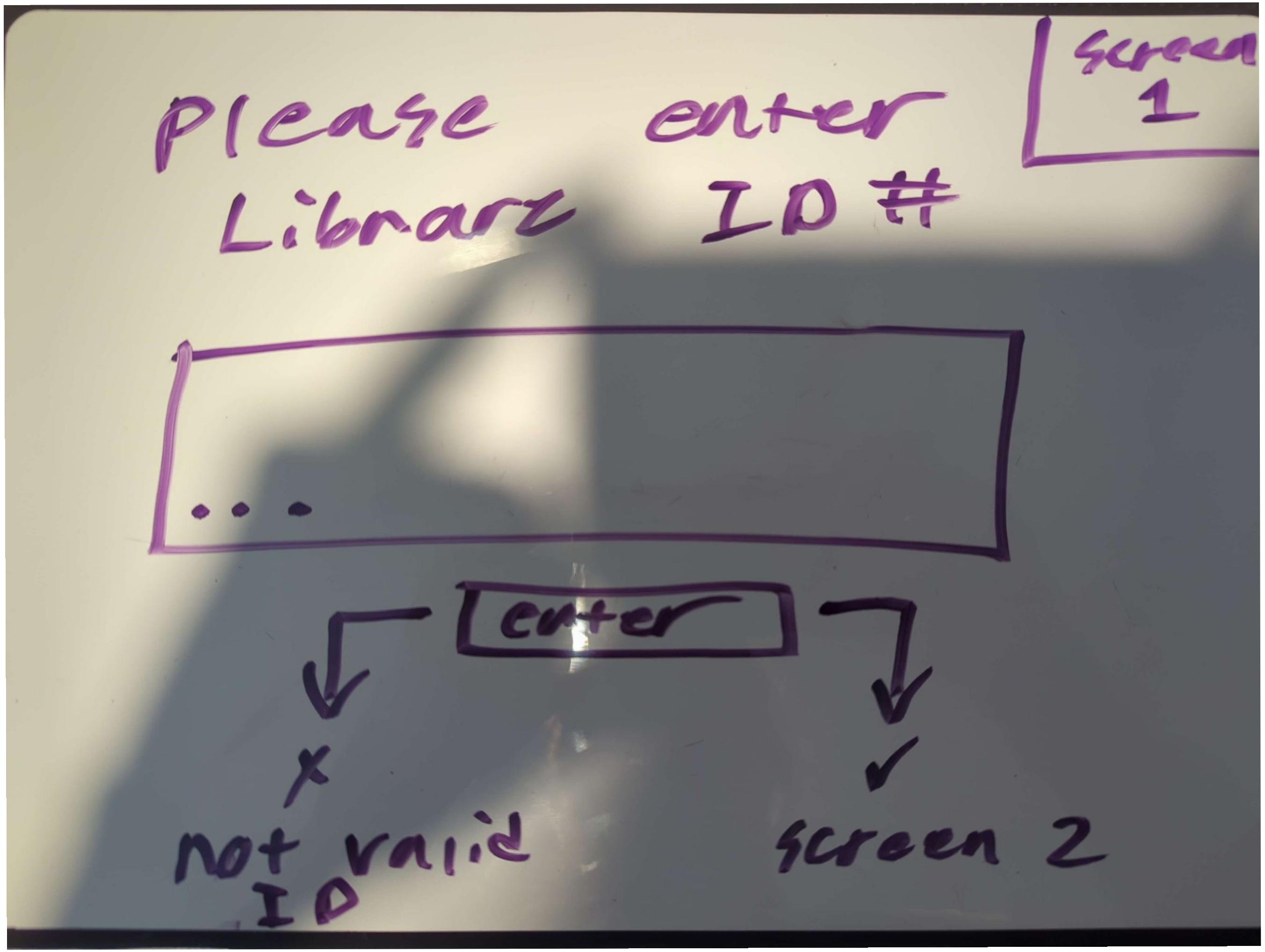
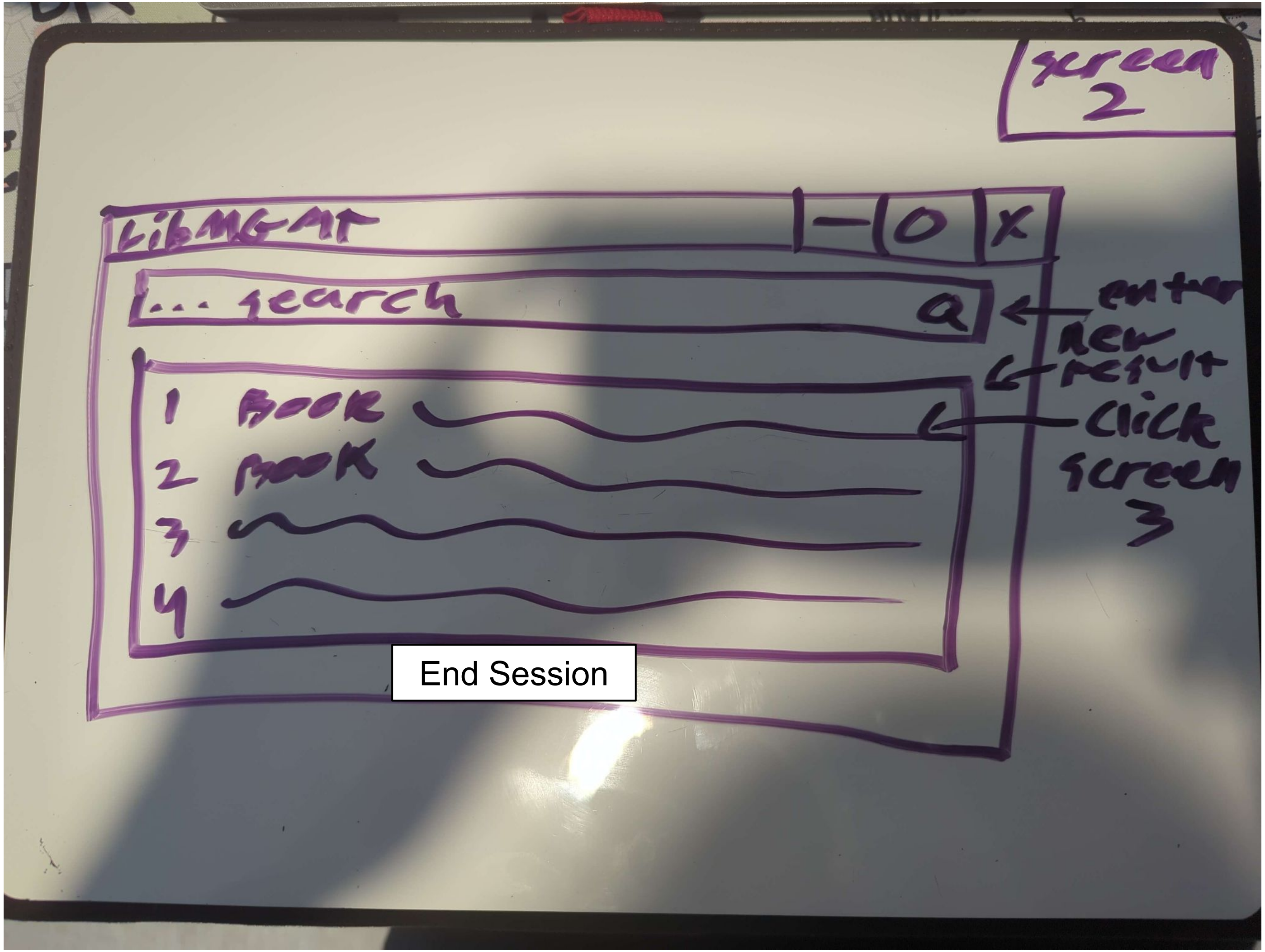
This project is a library kiosk application that allows users to check out and return books. Users can search for books by entering a keyword into the search bar and selecting a search option such as title, author, or genre. It requires a valid user ID before allowing access to the search screen. When a user selects a book to check out or return, a valid employee ID is required to confirm the action. This is to ensure the accuracy of the availability status of each book.



Screen 1:
Requests user ID Number to continue to Screen 2.

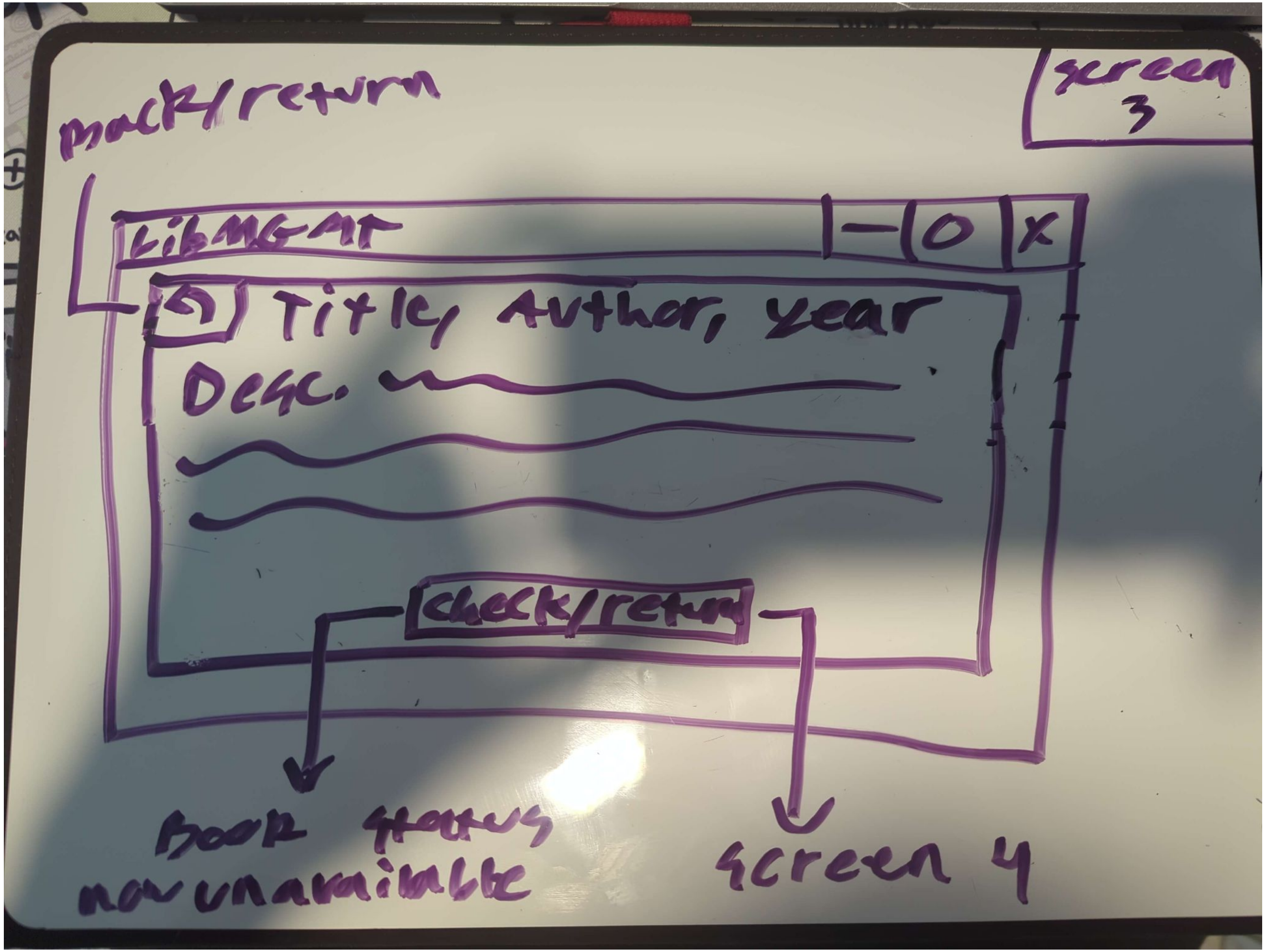
When an invalid ID is entered, the text prompt appears: “A valid user ID is required.”



Screen 2:
Shows the search bar with search options (author, title, genre).

The search results are clickable and take you to Screen 3.

Button: “End Session” — returns to Screen 1.

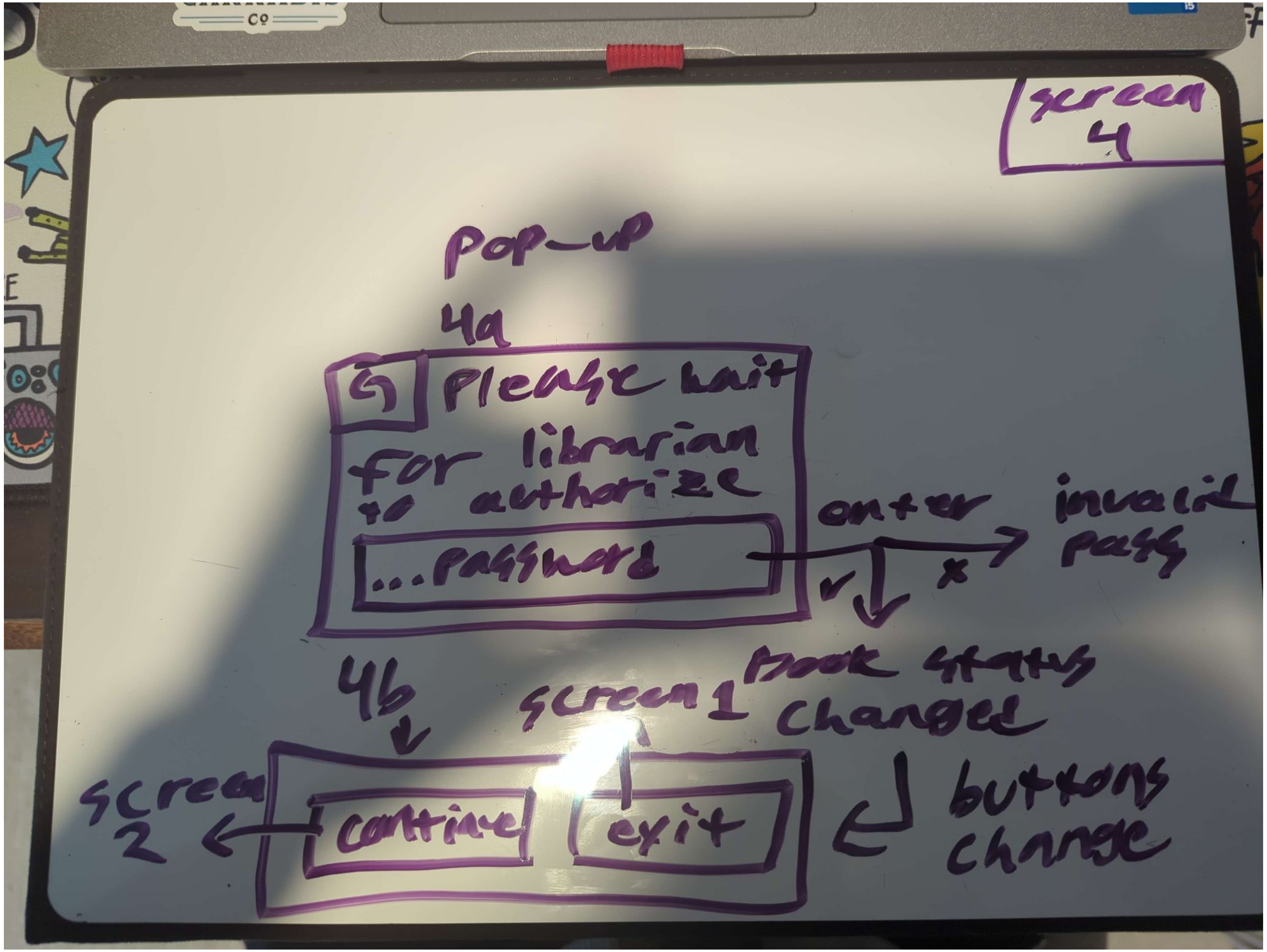


Screen 3:
This page shows the book's info (Title, Author, Genre, Publish Year, Book ID, Availability).

Button1: “Back to Search” or left-arrow icon

Button2:
if available: “Checkout”-- Screen 4 appears

if unavailable: “Return”-- Screen 4 appears



4th Screen: Confirmation Popup

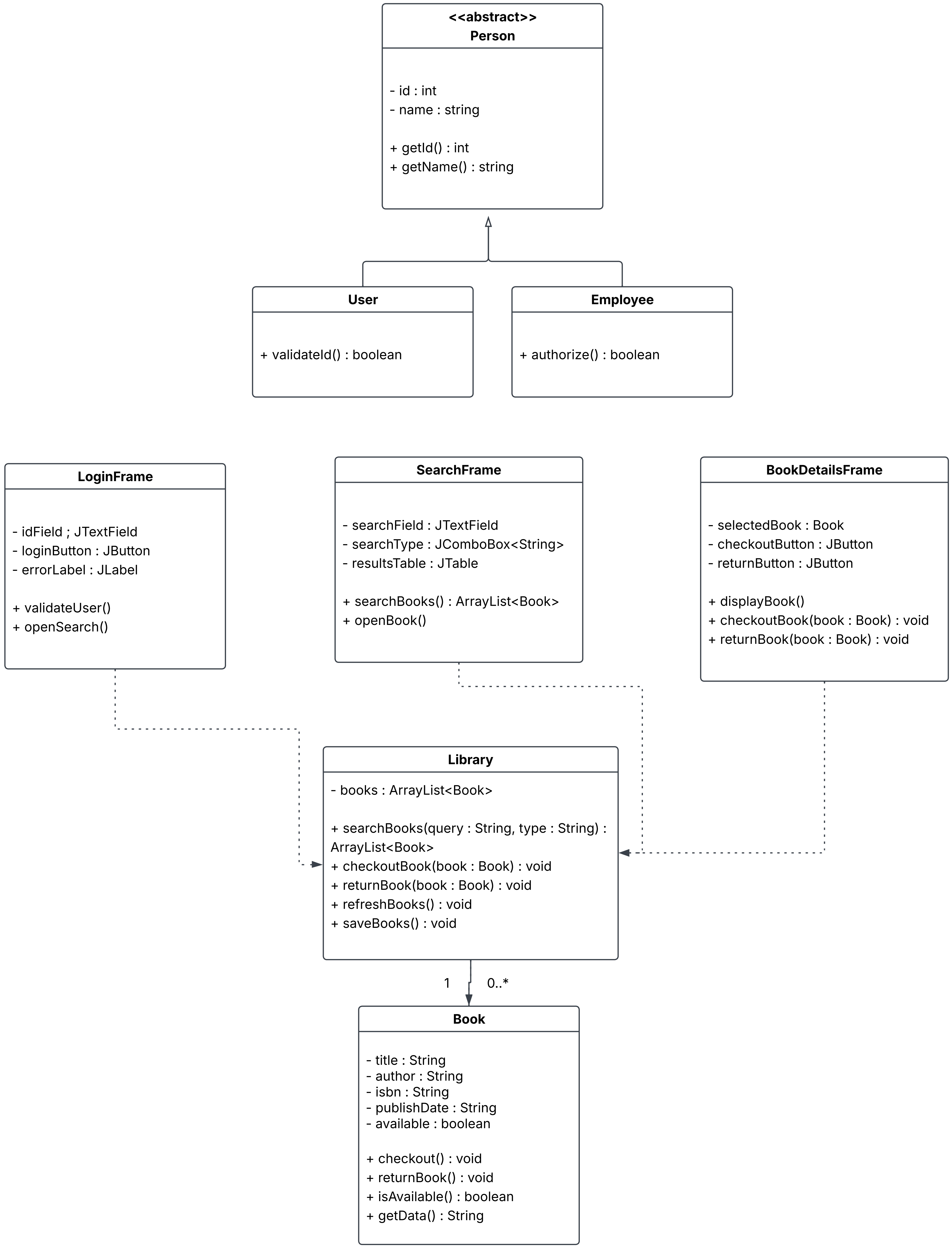
When a book is checked out or returned, a popup requests a valid employee ID.

Buttons:
Enter — submits the employee ID.
Cancel — returns to Screen 3.

When input is entered, the following text and buttons appear:

If valid:
"Database Updated!"
Button: OK

If invalid:
"A valid employee ID is required."
Button1: "Continue" — returns to Screen 2.
Button2: "End Session" — returns to Screen 1.



<<abstract>>
Person
- id : int
- name : string
+ getId() : int
+ getName() : string
User
+ validateId() : boolean
Employee
+ authorize() : boolean
LoginFrame
- idField ; JTextField
- loginButton : JButton
- errorLabel : JLabel
+ validateUser()
+ openSearch()
SearchFrame
- searchField : JTextField
- searchType : JComboBox<String>
- resultsTable : JTable
+ searchBooks() : ArrayList<Book>
+ openBook()
BookDetailsFrame
- selectedBook : Book
- checkoutButton : JButton
- returnButton : JButton
+ displayBook()
+ checkoutBook(book : Book) : void
+ returnBook(book : Book) : void
Library
- books : ArrayList<Book>
+ searchBooks(query : String, type : String) : ArrayList<Book>
+ checkoutBook(book : Book) : void
+ returnBook(book : Book) : void
+ refreshBooks() : void
+ saveBooks() : void
1
0..*
Book
- title : String
- author : String
- isbn : String
- publishDate : String
- available : boolean
+ checkout() : void
+ returnBook() : void
+ isAvailable() : boolean
+ getData() : String

E. Describe the Teamwork | Collaboration

To describe the teamwork and collaboration, answer the following questions:

I. How often did your team meet on this part of the assignment and how much time did your team spend approximately in total?
II. Did your team clearly divide up the work between team members and set clear timelines when individual work needed to be completed?
III. Were team members responsive and respectful of other people's ideas?
IV. What was the most challenging part for your team?
V. Briefly describe the work distribution. Did each team member contribute to the design in a significant way?
VI. Include the pebble distribution.
The **pebble distribution** (https://slcc.instructure.com/courses/1272217/pages/pebble-distribution) should have been discussed with the team before including it in the video. If the pebble distribution is uneven, describe the reason why.

6e1: We had 2 meetings, and spent about 4 hours total in meeting. We all spent an hour each on our own working on our assigned tasks and familiarizing ourselves with both the assignment and git/github for future collaboration. For 15 hours of total labor.

6e2: Ty Bride created the first draft of the UML diagram, while Montana and Morgan created the prototype. Montana focused writing the descriptions and requirements, while Morgan created the physical prototype. All team members contributed at least a little bit to all aspects of this assignment.

6e3: All team members were respectful gentlemen who maintained frequent contact with other team members, and respectfully communicated ideas, thoughts, and requirements during the development of this phase.

6e4: creating the UML and ensuring it was up to standards took a decent amount of time. The brainstorming session was relatively simple.

6e5: While all team members contributed in some part to all aspects of the assignment, we did have separation of tasks. Montana helped brainstorm and pitched the original idea for the library kiosk, and also wrote the verbal description of what we are looking to create. Morgan created the prototype. Ty created the UML diagram.

6e6: 33-33-33, all team members contributed evenly. I got hungry and ate the remaining 1 pebble.